1. Synthetic-only: design and factors

Research question: What choices does the experiment vary?

Method: Analyze this training dataset independently. The factorial grid is 5 datasets x 2 generators x 2 has_y values x 3 labelers x 2 seeds = 120 macro-F1 scores.

Conclusion: All relabeled combinations are observed. The 20 generated-target runs and 10 real-only classification runs remain reference comparisons, outside this factorial ANOVA.

Factor	Meaning	Levels
dataset	Real-data prediction task.	Adult; Census KDD; Covertype; MNIST12; MNIST28.
generator	Model fitted on real training data to generate synthetic rows.	CTGAN; TVAE.
has_y	Whether the generator is fitted with target Y.	0: fit X only. 1: fit (X,Y).
labeler	Predictor fitted on real training data to replace synthetic targets.	RF: random forest. XGB: XGBoost. DNN: neural network.
seed	Categorical experiment random-number setting.	42; 43. This is not a continuous covariate.

Reference	Construction	Role
Generated targets	Fit (X,Y), retain generated Y. No labeler.	Two generators x two seeds x five tasks = 20 scores.
Real only	Train downstream model on original real training rows.	Two seeds x five tasks = 10 scores.

Training rows: 100000 synthetic rows only. Mix concatenates all rows; it is not a fixed 50/50 mixture. Both reports use the same experimental scope, with separate statistical fits.

X = features; **Y** = target. has_y describes generator fitting, not whether the final training table contains Y. Every relabeled table contains X and predicted Y. Generated targets have no has_y=0 case and no labeler; that nonexistent combination is never estimated.

Macro F1 is the equal-weight average of F1 across target classes, on a 0-1 scale. The downstream evaluator is a dataset-specific neural model. RF/XGB/DNN are labelers. News and Housing regression scores are analyzed separately on page 5.

2. ANOVA: which factors affect macro F1?

Research question: Do generator, has_y, labeler, seed or their interactions change relabeled-data utility?

Method: Type II ANOVA: macro_F1 ~ dataset + generator * has_y * labeler * seed. All five factors are categorical; sum-to-zero coding is used.

Conclusion: Labeler differs (nominal p = 1.03e-05). Generator and has_y have small average effects; seed and treatment interactions are nonsignificant in this model.

```
macro_F1 ~ dataset + generator * has_y * labeler * seed
```

Term	SS	df	F	Nominal p	Sig.
dataset	**2.1890**	**4**	**266.10**	**1.17e-49**	***
generator	0.0002	1	0.08	0.7746	
has_y	0.0004	1	0.19	0.6671	
labeler	**0.0536**	**2**	**13.04**	**1.03e-05**	***
seed	3.77e-05	1	0.02	0.8926	
generator x has_y	1.99e-05	1	0.01	0.9219	
generator x labeler	0.0001	2	0.03	0.9667	
has_y x labeler	0.0001	2	0.02	0.9813	
generator x seed	0.0006	1	0.31	0.5790	
has_y x seed	0.0003	1	0.13	0.7227	
labeler x seed	0.0003	2	0.07	0.9348	
generator x has_y x labeler	0.0001	2	0.03	0.9667	
generator x has_y x seed	3.61e-05	1	0.02	0.8949	
generator x labeler x seed	0.0007	2	0.17	0.8406	
has_y x labeler x seed	0.0006	2	0.16	0.8555	
generator x has_y x labeler x seed	0.0001	2	0.02	0.9844	
Residual	0.1892	92			

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal Type II ANOVA p.

* includes all main effects, two-way, three-way and four-way interactions among generator, has_y, labeler and seed. For example, has_y x labeler asks whether including Y helps RF, XGB and DNN differently. Dataset is an additive block in this equation.

Marginal comparison	Mean F1 change	95% model CI
TVAE - CTGAN	+0.0024	[-0.0141, +0.0188]
has_y=1 - has_y=0	+0.0036	[-0.0129, +0.0200]
Seed 43 - seed 42	-0.0011	[-0.0176, +0.0153]

SS = sum of squares; df = numerator degrees of freedom; F = term mean square / residual mean square. Residual df = 92. Intervals use the ordinary model assumptions and are unadjusted. Shared test partitions make nominal ANOVA inference exploratory.

3. Does replacing generated targets improve utility?

Research question: Do RF/XGB/DNN targets outperform the generator's own targets, and does has_y matter?

Method: Match within dataset, seed and generator. delta[d] = mean(s,g,labeler) F1_relabel[d] - mean(s,g) F1_generated[d]. Test five dataset differences with t = mean(delta)/(SD(delta)/sqrt(5)), df=4.

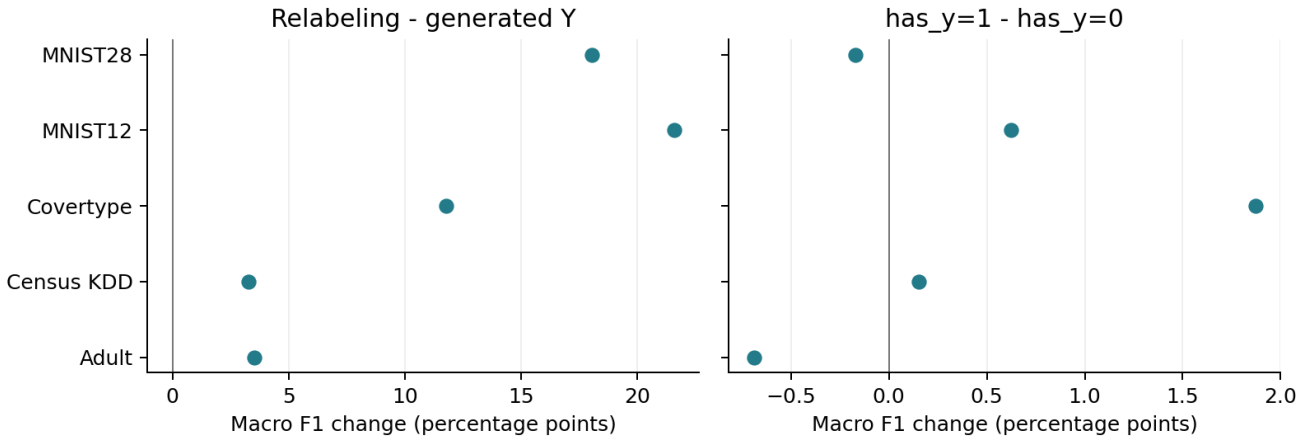
Conclusion: Relabeling gains **+0.1163 macro F1 (17.38% relative)**. has_y=1 minus 0 is **+0.0036**. Neither main comparison passes Holm-adjusted p < .05 across tasks.

Comparison	First mean	Second mean	F1 change	Relative
Relabeled - generated	0.7855	0.6692	+0.1163	17.38%
has_y=1 - has_y=0	0.7873	0.7837	+0.0036	0.46%
has_y=1 - generated	0.7873	0.6692	+0.1181	17.65%
has_y=0 - generated	0.7837	0.6692	+0.1145	17.11%

Comparison	95% paired-t CI	Holm p	Sig.
Relabeled - generated	[+0.0133, +0.2193]	0.1397	
has_y=1 - has_y=0	[-0.0085, +0.0156]	0.4575	
has_y=1 - generated	[+0.0132, +0.2230]	0.1397	
has_y=0 - generated	[+0.0132, +0.2159]	0.1397	

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses Holm-adjusted dataset-level p over four planned contrasts.

Why the gain misses Holm significance: the five task gains range from 3.28 to 21.58 percentage points; SE = 3.71 points. Raw p = 0.0350; Holm p = 0.1397. The smallest of four raw p-values is 0.0349, above the first Holm cutoff .05/4 = .0125. None passes. Large gains do not imply precise estimates.



Five dataset means show the main contrasts. Positive values favor relabeling or has_y=1.

All datasets, seeds and generators receive equal weight; relabeling also averages the three labelers. has_y compares only relabeled runs. The generated baseline is matched once per dataset/seed/generator and is not treated as six independent copies.

Relative gain = F1 change / second mean; .01 F1 is one percentage point. Confidence intervals are unadjusted. Exact sign-flip results are supplied; with five tasks the minimum two-sided p is .0625. This is not evidence that every future task improves.

4. How large is each labeler's improvement?

Research question: For each generator and has_y, how much do RF, XGB and DNN improve on generated targets?

Method: delta[d] = mean(s) F1_relabel[d,g,has_y,labeler] - mean(s) F1_generated[d,g]. Paired t-test across five datasets, df=4; Holm over twelve comparisons.

Conclusion: Largest descriptive gain: CTGAN / DNN / has_y=1 (**+0.2287**). 0 of twelve baseline comparisons pass Holm-adjusted p < .05.

Generator	Labeler	has_y	F1 mean	F1 gain	Relative	Holm p	Sig.
CTGAN	RF	0	0.7670	+0.1778	30.18%	0.6281	
CTGAN	XGB	0	0.7664	+0.1772	30.08%	0.6281	
CTGAN	DNN	0	0.8130	+0.2239	37.99%	0.5217	
CTGAN	RF	1	0.7666	+0.1774	30.11%	0.6281	
CTGAN	XGB	1	0.7752	+0.1860	31.56%	0.6281	
CTGAN	DNN	1	0.8179	+0.2287	38.82%	0.5217	
TVAE	RF	0	0.7676	+0.0184	2.45%	0.6281	
TVAE	XGB	0	0.7746	+0.0254	3.38%	0.3208	
TVAE	DNN	0	0.8138	+0.0645	8.61%	0.6281	
TVAE	RF	1	0.7711	+0.0219	2.92%	0.5687	
TVAE	XGB	1	0.7768	+0.0275	3.67%	0.1625	
TVAE	DNN	1	0.8164	+0.0672	8.97%	0.6281	

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses Holm-adjusted dataset-level p across twelve baseline comparisons.

Matched generated-target comparator	Mean macro F1
CTGAN	0.5892
TVAE	0.7492

These are comparisons against each generator's own generated targets, not pairwise tests between labelers. The labeler ANOVA on page 2 tests differences among RF/XGB/DNN. An observed largest gain does not establish a universally best labeler.

Means average seeds within tasks and then give each task equal weight. Per-task differences, confidence intervals, exact tests and fitted marginal means are in the supplied tables. Generated baselines remain outside the factorial fit.

5. Regression: what changes on News and Housing?

Research question: Does relabeling improve regression utility under the same constructions?

Method: Compare saved scores with generated targets. Means average seeds, generators and, for relabeled runs, RF/XGB/DNN. Pages 7-8 separately test factor effects within relabeled runs.

Conclusion: Housing relabeling improves all three average metrics. News relabeling improves R2 but worsens NMAE and D2. Baseline gains below are descriptive; ANOVA tests are exploratory.

```
R2 = 1 - sum((Y - prediction)^2) / sum((Y - mean(Y))^2)
```

```
NMAE = mean(abs(Y - prediction)) / SD_test(Y), ddof = 0
```

```
D2 = 1 - MAE_prediction / MAE_test_median_baseline
```

Dataset / metric	Real only	Generated Y	Relabeled has_y=0	Relabeled has_y=1
News / R2	-0.0290	-0.0235	0.0027	-0.0004
News / NMAE	0.2686	0.2546	0.2597	0.2587
News / D2	-0.0445	0.0099	-0.0098	-0.0060
Housing / R2	0.5125	0.6730	0.7946	0.7930
Housing / NMAE	0.5507	0.4146	0.3049	0.3022
Housing / D2	0.2822	0.4595	0.6025	0.6061

Relabeled - generated	R2 change	D2 change
News	+0.0246	-0.0178
Housing	+0.1208	+0.1448

Higher R2/D2 is better; lower NMAE is better. R2 measures squared error against the test-mean baseline; D2 measures absolute error against the test-median baseline. Negative R2/D2 means worse than that baseline.

News uses log-target reruns scored in original units; Housing uses no-Faker reruns. NMAE and D2 read saved test normalization and verified sidecars; denominators are not recomputed for this report. Real-only runs are references.

Scope: 406 verified selected records across both training analyses: 290 classification and 116 regression, plus 116 verified D2 sidecars. The 14 real-only runs are shared references. Credit, Intrusion, NB/PCA-GMM and simulated experiments are excluded.

MNIST caveat: output heads were repaired; retained MNIST12 TVAE synthetic tables still have documented holdout-feature collisions. Seeds share prepared test partitions. Neither seed supplies an independent new test sample.

6. Model checks: do common effects describe each task?

Research question: Do classification effects vary by task, and which regression factors matter within News or Housing?

Method: Use nested F tests for the classification equations below. For each regression dataset, fit R2 or D2 ~ seed + generator * has_y * labeler.

Conclusion: Classification dataset interactions are strong (p = 1.86e-09). Regression effects also depend on the task; use the within-task p-values below and signed changes on pages 7-8.

`Seed block: macro_F1 ~ dataset + seed + generator * has_y * labeler`

`Dataset check: macro_F1 ~ dataset * (has_y * labeler)`

`+ generator * has_y * labeler + seed`

Added terms	df num / den	F	Nominal p	Sig.
Seed interactions	11 / 92	0.116	0.9998	
Dataset interactions	**20 / 83**	**6.042**	**1.86e-09**	***

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal nested-model p; these are supporting checks.

`Within each regression dataset: R2 or D2 ~ seed + generator * has_y * labeler`

Dataset / outcome	Generator p	has_y p	Labeler p	Seed p
News / R2	**0.0118 ***	0.0704 .	**0.0093 ****	0.6229
Housing / R2	0.0540 .	0.8662	**0.0065 ****	0.1197
News / D2	**0.0001 *****	0.2442	0.1783	0.2980
Housing / D2	**0.0111 ***	0.1949	**3.31e-07 *****	0.1719

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal within-task ANOVA p; bold cells mark p < .05.

Each regression task has 24 relabeled scores, 13 coefficients and 11 residual df. A full generator * has_y * labeler * seed fit within one task would be saturated: 24 coefficients, zero residual df. The stated within-task equation keeps seed additive; its omitted seed interactions form the residual. Full tables are supplied.

Primary model: 120 scores, 28 identifiable coefficients, 92 residual df. Seed-block rank = 17; dataset-check rank = 37. All ANOVA F values were independently reproduced. Primary residual Shapiro p = 0.0033; adjusted R-squared = 0.8995.

The ordinary ANOVA assumes independent errors with common variance. Shared test sets, nonnormal residuals and dataset-specific treatment effects limit its p-values; page 3 therefore reports paired evidence across five dataset means. Seed is fixed at 42/43, not a sampled population of seeds.

Provenance: original score snapshot Oct 10, 2026, 1:08:42 a.m. Chicago; all 522 source/sidecar hashes rechecked Oct 10, 2026 at 02:53 PM Chicago. Generator settings remain 500 epochs and 100,000 rows. Census selects checkpoints by weighted macro-F1; others use development loss. Experiments were not rerun.

7. Regression ANOVA: which factors affect R2?

Research question: Among relabeled runs, do generator, has_y, labeler, seed or their interactions change R2?

Method: Type II ANOVA: R2 ~ dataset + generator * has_y * labeler * seed. All factors are categorical with sum-to-zero coding; synthetic and mix are fitted independently.

Conclusion: No treatment main effect or interaction passes nominal p < .05 in this common-effects model. has_y has no detected common effect; a significant term means choices differ, not that every choice improves utility.

```
R2 ~ dataset + generator * has_y * labeler * seed
```

Term	SS	df	F	Nominal p	Partial eta2	Sig.
dataset	**7.5400**	**1**	**13941.79**	**1.62e-33**	**0.998**	*******
generator	0.0006	1	1.15	0.2947	0.048	
has_y	0.0001	1	0.12	0.7332	0.005	
labeler	0.0029	2	2.71	0.0879	0.191	.
seed	0.0006	1	1.11	0.3036	0.046	
generator x has_y	2.64e-06	1	4.89e-03	0.9449	2.12e-04	
generator x labeler	0.0012	2	1.13	0.3392	0.090	
has_y x labeler	4.31e-05	2	0.04	0.9610	0.003	
generator x seed	0.0003	1	0.59	0.4499	0.025	
has_y x seed	2.99e-05	1	0.06	0.8161	0.002	
labeler x seed	0.0005	2	0.50	0.6145	0.041	
generator x has_y x labeler	4.88e-05	2	0.05	0.9560	0.004	
generator x has_y x seed	2.72e-06	1	0.01	0.9441	2.19e-04	
generator x labeler x seed	0.0011	2	1.06	0.3633	0.084	
has_y x labeler x seed	0.0001	2	0.09	0.9185	0.007	
generator x has_y x labeler x seed	0.0001	2	0.12	0.8881	0.010	
Residual	0.0124	23				

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal exploratory ANOVA p; no multiplicity adjustment.

Marginal comparison	Pooled change	News change	Housing change
TVAE - CTGAN	-0.0072	+0.0047	-0.0191
has_y=1 - has_y=0	-0.0023	-0.0031	-0.0015
Seed 43 - seed 42	+0.0071	-0.0008	+0.0149
RF - DNN	+0.0018	-0.0053	+0.0088
XGB - DNN	+0.0174	-0.0070	+0.0417

Positive changes favor the first choice. Pooled changes give the two tasks equal weight. Within-task model p-values are on page 6; unadjusted 95% model intervals and contrast p-values are supplied in regression_factor_effects.csv.

48 observed scores; 25 identifiable coefficients; 23 residual df. Partial eta2 = SS_term / (SS_term + SS_residual). These exploratory tests use two fixed datasets and shared test partitions. They do not test relabeled versus generated targets; those baseline changes are on page 5. Dataset SS reflects different task score levels.

8. Regression ANOVA: which factors affect D2?

Research question: Among relabeled runs, do generator, has_y, labeler, seed or their interactions change D2?

Method: Type II ANOVA: D2 ~ dataset + generator * has_y * labeler * seed. All factors are categorical with sum-to-zero coding; synthetic and mix are fitted independently.

Conclusion: Nominal main effects: generator (p = 0.0164), labeler (p = 0.0475). has_y has no detected common effect; a significant term means choices differ, not that every choice improves utility.

```
D2 ~ dataset + generator * has_y * labeler * seed
```

Term	SS	df	F	Nominal p	Partial eta2	Sig.
dataset	**4.4977**	**1**	**15390.72**	**5.20e-34**	**0.999**	*******
generator	**0.0020**	**1**	**6.70**	**0.0164**	**0.226**	*****
has_y	0.0002	1	0.56	0.4608	0.024	
labeler	**0.0020**	**2**	**3.49**	**0.0475**	**0.233**	*****
seed	0.0002	1	0.53	0.4740	0.023	
generator x has_y	4.39e-05	1	0.15	0.7020	0.006	
generator x labeler	0.0003	2	0.59	0.5632	0.049	
has_y x labeler	0.0006	2	0.96	0.3987	0.077	
generator x seed	0.0002	1	0.69	0.4151	0.029	
has_y x seed	0.0001	1	0.21	0.6481	0.009	
labeler x seed	9.28e-06	2	0.02	0.9843	0.001	
generator x has_y x labeler	0.0001	2	0.14	0.8670	0.012	
generator x has_y x seed	4.59e-06	1	0.02	0.9014	0.001	
generator x labeler x seed	3.32e-05	2	0.06	0.9449	0.005	
has_y x labeler x seed	0.0001	2	0.15	0.8652	0.013	
generator x has_y x labeler x seed	7.79e-06	2	0.01	0.9868	0.001	
Residual	0.0067	23				

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal exploratory ANOVA p; no multiplicity adjustment.

Marginal comparison	Pooled change	News change	Housing change
TVAE - CTGAN	-0.0128	-0.0176	-0.0079
has_y=1 - has_y=0	+0.0037	+0.0038	+0.0036
Seed 43 - seed 42	+0.0036	+0.0034	+0.0038
RF - DNN	-0.0075	+0.0046	-0.0196
XGB - DNN	+0.0085	-0.0030	+0.0199

Positive changes favor the first choice. Pooled changes give the two tasks equal weight. Within-task model p-values are on page 6; unadjusted 95% model intervals and contrast p-values are supplied in regression_factor_effects.csv.

48 observed scores; 25 identifiable coefficients; 23 residual df. Partial eta2 = SS_term / (SS_term + SS_residual). These exploratory tests use two fixed datasets and shared test partitions. They do not test relabeled versus generated targets; those baseline changes are on page 5. Dataset SS reflects different task score levels.